بسم الله الرحمن الرحيم

Physics II (EBA1104 & BA114)

Section

Temperature

1. Convert the following temperatures to their values on the Fahrenheit and Kelvin scales:

(a) the sublimation point of dry ice −78.5 °C.

(b) human body temperature, 37 °C.

a) $$T_F = \left(\frac{9}{5}\right)T_C + 32 = \left(\frac{9}{5}\right) \times -78.5 + 32 = -109.3\ °F$$

$$T_K = T_C + 273 = -78.5 + 273 = 194.5\ K$$

b) $$T_F = \left(\frac{9}{5}\right)T_C + 32 = \left(\frac{9}{5}\right) \times 37 + 32 = 98.6\ °F$$

$$T_K = T_C + 273 = 37 + 273 = 310\ K$$

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2. The boiling point of liquid hydrogen is 20.3 K at atmospheric pressure. What is this temperature on

(a) the Celsius scale

(b) the Fahrenheit scale

$$T_K = T_C + 273 \longrightarrow 20.3 = T_C + 273 \longrightarrow T_C = -252.7\ °C$$

$$T_F = \left(\frac{9}{5}\right)T_C + 32 = \left(\frac{9}{5}\right) \times -252.7 + 32 = -422.86\ °F$$

Thermal expansion in solids and liquids

3. A pair of eyeglass frames is made of epoxy plastic. At room temperature (20 °C), the frames have circular lens holes 2.20 cm in radius. To what temperature must the frames be heated if lenses 2.21 cm in radius are to be inserted in them? The average coefficient of linear expansion for epoxy is 1.30×10^{-4} °C^{-1}

$L_i = 2.2\ \text{cm} = 2.2\times10^{-2}\ \text{m}$ $\qquad T_i = 20\ °\text{C}$

$L_F = 2.21\ \text{cm} = 2.21\times10^{-2}\ \text{m}$

$$\Delta L = \alpha L_i\ \Delta T \longrightarrow \left(L_F - L_i\right) = \alpha L_i\left(T_F - T_i\right)$$

$$\left(2.21\times10^{-2} - 2.2\times10^{-2}\right) = 1.3\times10^{-4}\times2.2\times10^{-2}\ \left(T_F - 20\right)$$

$$T_F = 54.96\ °\text{C}$$

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4. The Trans-Alaska pipeline is 1300 km long, reaching from Prudhoe Bay to the port of Valdez. It experiences temperatures from −73 °C to +35 °C. How much does the steel pipeline expand because of the difference in temperature? How can this expansion be compensated for? [α for steel = 11×10^{-4} °C^{-1}]

$L_i = 1300\ \text{Km} = 1300\times10^3\ \text{m}$ $\qquad T_i = -73\ °\text{C}$ $\qquad T_F = 35\ °\text{C}$

$$\Delta L = \alpha L_i\ \Delta T \longrightarrow \Delta L = \alpha L_i\left(T_F - T_i\right)$$

$$\Delta L = 11\times10^{-6}\times1300\times10^3\ \left(35-(-73)\right) = 1.54\times10^3\ \text{m}$$

5. The active element of a certain laser is made of a glass rod 30.0 cm long and 1.50 cm in diameter. Assume the average coefficient of linear expansion of the glass is equal to $9.00 \times 10^{-6}\ {}^{o}C^{-1}$. If the temperature of the rod increases by 65 ^{o}C, what is the increase in (a) its length, (b) its diameter, and (c) its volume.

$$L_i = 30\ \text{cm} = 30\times10^{-2}\,\text{m} \qquad D_i = 1.5\ \text{cm} = 1.5\times10^{-2}\,\text{m} \qquad \Delta T = 65\ {}^{\circ}\text{C}$$

$$\alpha\ for\ \text{glass} = 9x\,10^{-6}\,{}^{o}C^{-1}$$

a) $$\Delta L = \alpha L_i\ \Delta T = 9\times10^{-6}\times30\times10^{-2}\times65 = 1.76\times10^{-4}\,\text{m}$$

b) $$\Delta D = \alpha D_i\ \Delta T = 9\times10^{-6}\times1.5\times10^{-2}\times65 = 8.78\times10^{-6}\,\text{m}$$

c) $$\Delta V = \beta V_i\ \Delta T$$

$$V_i = \pi r^2 L = \pi\left(\frac{1.5\times10^{-2}}{2}\right)^2\times30\times10^{-2} = 5.3\times10^{-5}\ \text{m}^3$$

$$\beta = 3\alpha\ =\ 3x\,9x\,10^{-6}\,{}^{o}C^{-1}$$

$$\Delta V = \beta V_i\ \Delta T = 3\times9\times10^{-6}\times5.3\times10^{-5}\times65 = 9.3\times10^{-8}\,\text{m}^3$$

6. The average coefficient of volume expansion for carbon tetrachloride is $5.81 \times 10^{-4}\ {}^{o}C^{-1}$. If a 50.0 gal steel container is filled completely with carbon tetrachloride when the temperature is 10.0 °C, how much will spill over when the temperature rises to 30.0 °C? [α for steel = $11 \times 10^{-6}\ {}^{o}C^{-1}$]. .

$$\beta = 5.81x\,10^{-4}\,{}^{o}C^{-1}$$

$$T_i = 10\ ^{\circ}C$$

$$V_i = 5\ \text{gal} = 50 \times 0.003785 = 0.189\ m^3$$

$$T_F = 30\ ^{\circ}C$$

$$\Delta V = \Delta V_C - \Delta V_S$$

$$\Delta V = \left(\beta_C - \beta_S\right) V_i\ \Delta T$$

$$\Delta V = \left(\beta_C - 3\alpha_S\right) V_i\ \left(T_F - T_i\right)$$

$$\Delta V = \left(5.81 \times 10^{-4} - 3 \times 11 \times 10^{-6}\right) \times 0.189 \times \left(30 - 10\right) = 2.07 \times 10^{-3}\ m^3$$

THANKS